

VentureMiner AI

API Specifications

Document: 08-Engineering / 19_api_specifications

Version: v1.0

Date: 2026-07-20

Author: VentureMiner AI Documentation Team

Status: Approved

Project: VentureMiner AI — AI Venture Intelligence Platform

Table of Contents

Document 19 — API Specifications

Table of Contents

1. Purpose & Scope
2. API principles
3. Versioning
4. Authentication
 - 4.1 Token scopes
5. Rate limits
6. Pagination
7. Idempotency
8. Errors
9. Webhooks
 - 9.1 Event types (excerpt)
10. Resource models
 - 10.1 Opportunity
 - 10.2 Validation run
11. Endpoint catalog (selected)
12. Deprecation policy
13. Appendix
 - 13.1 Revision history
 - 13.2 Cross-references

Document 19 — API Specifications

The contract for every public REST endpoint. The OpenAPI 3.1 spec is the source of truth; this document is the human-readable companion.

Table of Contents

1. Purpose & Scope
2. API principles
3. Versioning
4. Authentication
5. Rate limits
6. Pagination
7. Idempotency
8. Errors
9. Webhooks
10. Resource models
11. Endpoint catalog
12. Deprecation policy
13. Appendix

1. Purpose & Scope

This document is the API contract. It defines how external clients (and internal services that act as clients) interact with the platform. The OpenAPI spec lives in `docs/api/openapi.yaml`; this document explains its structure and the rules it must follow.

2. API principles

- RESTful with pragmatic departures (e.g. RPC-style action endpoints for long-running tasks).
- JSON request and response bodies.
- UTF-8 encoding.
- ISO-8601 timestamps.
- camelCase JSON keys; snake_case query and path parameters.
- Versioned URL prefix (`/v1/`).
- Predictable resource shapes — the same shape returned by GET, POST, PATCH where possible.
- Cursor pagination for lists (stable under inserts).

3. Versioning

- Breaking changes include: removing a field, renaming a field, changing a field type, changing a status enum value, removing an endpoint, changing an auth flow.
- The v1 API is supported for 12 months after v2 GA.

4. Authentication

- OAuth 2.1 with PKCE for browser flows.
- API tokens for server-to-server.
- All endpoints require a valid bearer token except /healthz, /readyz, /docs, /redoc, and the marketing site.

4.1 Token scopes

Tokens carry scopes. Examples:

- opportunity:read
- opportunity:write
- validation:read
- validation:write
- report:read
- report:write
- rubric:read
- rubric:write
- workspace:read
- workspace:write
- member:read
- member:write
- webhook:read
- webhook:write

A token is rejected with 403 if a required scope is missing.

5. Rate limits

Free	60	1,000	5,000
Solo	300	10,000	50,000
Team	1,500	50,000	250,000
Enterprise	Custom	Custom	Custom

- 429 returned with Retry-After.

- ?cursor=...&limit;=...
- Response: { "items": [...], "next_cursor": "..." | null }.
- limit capped at 100 (default 25).
- Stable order: created_at DESC, id ASC.

7. Idempotency

- Mutating endpoints accept an Idempotency-Key header.
- The server stores the result for 24h.
- Retries with the same key return the original result (or a 409 if the request body differs).

8. Errors

All errors follow the same envelope:

```
{
  "error": {
    "code": "string",
    "message": "human-readable",
    "details": { },
    "trace_id": "..."
  }
}
```

400	validation_error	Missing required field
401	unauthenticated	Token missing or expired
403	forbidden	Missing scope
404	not_found	Resource not found
409	conflict	Idempotency key reused with different body
422	unprocessable	Business rule violation
429	rate_limited	Slow down
500	internal_error	Try again; trace_id is the support token
503	service_unavailable	Retry with backoff

9. Webhooks

- Outbound webhooks are signed (X-Signature: sha256=...).
- Events use CloudEvents 1.0 envelope.

9.1 Event types (excerpt)

- opportunity.created
- opportunity.updated
- opportunity.archived
- validation.requested
- validation.completed
- validation.failed
- score.computed
- report.generated
- report.exported
- subscription.changed

10. Resource models

The complete JSON schemas are in the OpenAPI spec. Selected highlights:

10.1 Opportunity

```
{
  "id": "uuid",
  "workspace_id": "uuid",
  "title": "string",
  "summary": "string",
  "status": "watching|validating|active|archived",
  "score": { "total": 7.4, "confidence": "high" },
  "tags": ["string"],
  "created_at": "ISO-8601",
  "updated_at": "ISO-8601"
}
```

10.2 Validation run

```
{
  "id": "uuid",
  "opportunity_id": "uuid",
  "depth": "quick|standard|deep",
  "status": "queued|running|succeeded|partial|failed",
  "started_at": "ISO-8601",
  "completed_at": "ISO-8601 | null"
}
```

POST	/v1/opportunities	Create an opportunity
GET	/v1/opportunities	List opportunities
GET	/v1/opportunities/{id}	Retrieve an opportunity
PATCH	/v1/opportunities/{id}	Update an opportunity
DELETE	/v1/opportunities/{id}	Archive (soft delete)
POST	/v1/opportunities/{id}/validate	Start a validation run
GET	/v1/validations/{id}	Retrieve a validation run
GET	/v1/validations/{id}/evidence	List evidence
GET	/v1/opportunities/{id}/score	Retrieve latest score
GET	/v1/opportunities/{id}/scores	Score history
POST	/v1/opportunities/{id}/reports	Generate a report
GET	/v1/reports/{id}	Retrieve a report
GET	/v1/reports/{id}.{format}	Export (pdf, docx, md, html)
GET	/v1/rubrics	List rubrics
POST	/v1/rubrics	Create a rubric
PATCH	/v1/rubrics/{id}/versions	New version
POST	/v1/discovery/runs	Start a discovery run
GET	/v1/discovery/runs/{id}	Retrieve a run
GET	/v1/workspaces	List workspaces
POST	/v1/workspaces	Create
GET	/v1/workspaces/{id}/members	List members
POST	/v1/workspaces/{id}/members	Invite
POST	/v1/webhooks	Register a webhook
GET	/v1/webhooks	List

12. Deprecation policy

- An endpoint can be marked Deprecated without removal.
- The response includes a Deprecation and Sunset header.
- Documentation shows a banner.
- 12-month notice before removal.
- A stable, deprecation-aware client is shipped in our SDK.

13.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

13.2 Cross-references

- TRD: Document 02.
- Backend: Document 05.
- Security: Document 21.

End of Document 19 — API Specifications. The OpenAPI spec is the source of truth; this document is the narrative.